

Math 1232 Summer 2025

Mastery Quiz 4

Due Tuesday, July 15

This mastery quiz has three topics. Everyone should do M2 and S3, but if you already have a 4/4 on M1, feel free to skip it. Please do your best, but don't worry if you make a minor error—your goal is to demonstrate your mastery of the underlying material.

Feel free to consult your notes, but please don't discuss the actual quiz questions with other students in the course.

Remember that you are trying to demonstrate that you understand the concepts involved. For all these problems, justify your answers and explain how you reached them. Do not just write “yes” or “no” or give a single number.

Please turn in this quiz in class on Tuesday. You may print this document out and write on it, or you may submit your work on separate paper; in either case make sure your name is clearly on it. If you absolutely cannot turn it in in person, you can submit it electronically but this should be a last resort.

Topics on this quiz:

- Major Topic 1: Transcendental Functions
- Major Topic 2: Integration Techniques
- Secondary Topic 3: Improper Integrals

Name: Solutions

Major Topic 1: Transcendental Functions

(a) Compute $\int \frac{1}{s \log_3(s)} ds$.

Solution. Set $u = \log_3(s)$ so $du = \frac{1}{s \ln(3)} ds$, and we have

$$\int \frac{1}{s \log_3(s)} ds = \int \frac{\ln(3) du}{u} = \ln(3) \ln |u| + C = \ln(3) \ln |\log_3(s)| + C.$$

(b) Compute $\frac{d}{dx} (\arctan(x^2))^2$.

Solution.

$$\frac{d}{dx} (\arctan(x^2))^2 = 2 \arctan(x^2) \frac{1}{1 + (x^2)^2} \cdot 2x = \frac{4x \arctan(x^2)}{1 + x^4}.$$

(c) Compute $\frac{d}{dx} \sin(x)^{\tan(x)}$.

Solution.

$$\begin{aligned} \ln y &= \tan(x) \ln |\sin(x)| \\ \frac{y'}{y} &= \sec^2(x) \ln |\sin(x)| + \tan(x) \frac{\cos(x)}{\sin(x)} \\ &= \sec^2(x) \ln |\sin(x)| + 1 \\ y' &= y (1 + \sec^2(x) \ln |\sin(x)|) \\ &= \sin(x)^{\tan(x)} (1 + \sec^2(x) \ln |\sin(x)|). \end{aligned}$$

Major Topic 2: Integration Techniques

(a) Compute $\int \sec^4(3t) dt$.

Solution. This is an even power of secant, so we want to set up a $u = \tan(3t)$ substitution. To make the substitution work, we need to keep $du = 3\sec^2(3t)dt$ in the integrand, and then rewrite the other copies of secant in terms of tangent using the $\tan^2(x) + 1 = \sec^2(x)$ identity.

$$\begin{aligned}\int \sec^4(3t) dt &= \int \sec^2(3t)(1 + \tan^2(3t)) dt \\&= \int \frac{1}{3}(1 + u^2) du \\&= \frac{u}{3} + \frac{u^3}{9} + C \\&= \frac{1}{9} (3 \tan(3t) + \tan^3(3t)) + C.\end{aligned}$$

(b) Compute $\int \frac{x^2 + x - 4}{(x + 3)^2(x + 1)} dx$.

Solution. This is a rational function, which is often a signal to use the partial fractions technique. Here we have the denominator already factored for us into three linear factors, so we'll need to try writing this as a sum of three different terms.

$$\begin{aligned}\frac{x^2 + x - 4}{(x + 3)^2(x + 1)} &= \frac{A}{x + 3} + \frac{B}{(x + 3)^2} + \frac{C}{x + 1} \\x^2 + x - 4 &= A(x + 3)(x + 1) + B(x + 1) + C(x + 3)^2 \\x = -3 : 2 &= -2B \Rightarrow B = -1 \\x = -1 : -4 &= 4C \Rightarrow C = -1 \\x = 0 : -4 &= 3A + B + 9C = 3A - 1 - 9 \Rightarrow A = 2 \\\frac{x^2 + x - 4}{(x + 3)^2(x + 1)} &= \frac{2}{x + 3} + \frac{-1}{(x + 3)^2} + \frac{-1}{x + 1} \\\int \frac{x^2 + x - 4}{(x + 3)^2(x + 1)} dx &= \int \frac{2}{x + 3} + \frac{-1}{(x + 3)^2} + \frac{-1}{x + 1} dx \\&= 2 \ln |x + 3| + \frac{1}{x + 3} - \ln |x + 1| + C.\end{aligned}$$

(c) Compute $\int_3^6 \frac{\sqrt{x^2 - 9}}{x} dx$.

Solution. We can take $x = 3 \sec(\theta)$, so that $dx = 3 \sec(\theta) \tan(\theta) d\theta$. Since this is a definite integral, we need to change the bounds, which means solving the equations $3 = 3 \sec(\theta)$ and $6 = 3 \sec(\theta)$ for θ . Since $\sec(\theta) = 1/\cos(\theta)$, we are solving $\cos(\theta) = 1$ and $\cos(\theta) = 1/2$, respectively.

$$\begin{aligned} \int_3^6 \frac{\sqrt{x^2 - 9}}{x} dx &= \int_0^{\pi/3} \frac{\sqrt{9 \sec^2(\theta) - 9}}{3 \sec(\theta)} \cdot 3 \sec(\theta) \tan(\theta) d\theta \\ &= \int_0^{\pi/3} \frac{3 \tan(\theta)}{3 \sec(\theta)} \cdot 3 \sec(\theta) \tan(\theta) d\theta \\ &= 3 \int_0^{\pi/3} \tan^2(\theta) d\theta = 3 \int_0^{\pi/3} \sec^2(\theta) - 1 d\theta \\ &= 3(\tan(\theta) - \theta) \Big|_0^{\pi/3} = 3\sqrt{3} - \pi. \end{aligned}$$

Secondary Topic 3: Improper Integrals

(a) Compute $\int_1^\infty \frac{\ln(3x)}{x^2} dx$.

Solution.

$$\begin{aligned}\int_1^\infty \frac{\ln(3x)}{x^2} dx &= \lim_{t \rightarrow \infty} \int_1^t \frac{\ln(3x)}{x^2} dx \\&= \lim_{t \rightarrow \infty} \left. \frac{-\ln(3x)}{x} \right|_1^t - \int_1^t \frac{-1}{x^2} dx \\&= \lim_{t \rightarrow \infty} \left. \frac{-\ln(3x)}{x} \right|_1^t - \left. \frac{1}{x} \right|_1^t \\&= \lim_{t \rightarrow \infty} \frac{-\ln(3t)}{t} + \frac{\ln(3)}{1} - \frac{1}{t} + 1 \\&= 1 + \ln(3) - \lim_{t \rightarrow \infty} \frac{\ln(3t)}{t} \\&\stackrel{\text{L'H}}{=} 1 + \ln(3) - \lim_{t \rightarrow \infty} \frac{\frac{1}{t}}{1} = 1 + \ln(3).\end{aligned}$$

(b) Compute $\int_1^3 \frac{x}{x^2 - 1} dx$.

Solution.

$$\begin{aligned}\int_1^3 \frac{x}{x^2 - 1} dx &= \lim_{t \rightarrow 1^+} \int_t^3 \frac{x}{x^2 - 1} dx \\&= \lim_{t \rightarrow 1^+} \frac{1}{2} \ln |x^2 - 1| \Big|_t^3 \\&= \lim_{t \rightarrow 1^+} \ln(8) - \ln |t^2 - 1| \\&= \ln(8) - \lim_{x \rightarrow 1^+} \ln |t^2 - 1| = \infty.\end{aligned}$$

So this integral doesn't converge.